DS MINI PROJECT :FLIGHT RESERVATION SYSTEM USING PYTHON

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# Flight Reservation System using Python
# Data Structures Mini Project

flights = [
    {"id": "F101", "source": "Pune", "destination": "Delhi", "fare": 4500, "seats": 3},
    {
        "id": "F102",
        "source": "Mumbai",
        "destination": "Bengaluru",
        "fare": 3500,
        "seats": 4,
    },
    {
        "id": "F103",
        "source": "Delhi",
        "destination": "Chennai",
        "fare": 5000,
        "seats": 2,
    },
    {"id": "F104", "source": "Pune", "destination": "Goa", "fare": 2500, "seats": 3},
]

# Dictionary stores bookings using booking ID as the key
bookings = {}

# Set ensures booking IDs remain unique
used_booking_ids = set()
next_booking_number = 1001
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def display_flights():
    print("\n--- AVAILABLE FLIGHTS ---")
    print("ID    Source      Destination   Fare   Seats")
    for flight in flights:
        print(
            f"{flight['id']:<6} "
            f"{flight['source']:<12} "
            f"{flight['destination']:<13} "
            f"Rs.{flight['fare']:<5} "
            f"{flight['seats']}"
        )


def search_flights():
    source = input("Enter source: ").strip().casefold()
    destination = input("Enter destination: ").strip().casefold()

    found = False
    for flight in flights:
        if (
            flight["source"].casefold() == source
            and flight["destination"].casefold() == destination
        ):
            print(flight)
            found = True

    if not found:
        print("No matching flight found.")
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def book_ticket():
    global next_booking_number

    flight_id = input("Enter flight ID: ").strip().upper()
    flight = next((f for f in flights if f["id"] == flight_id), None)

    if flight is None:
        print("Invalid flight ID.")
        return

    if flight["seats"] <= 0:
        print("Sorry, no seats are available.")
        return

    name = input("Enter passenger name: ").strip()
    if not name:
        print("Passenger name cannot be empty.")
        return

    # Generate a unique booking ID
    while f"B{next_booking_number}" in used_booking_ids:
        next_booking_number += 1

    booking_id = f"B{next_booking_number}"
    next_booking_number += 1

    bookings[booking_id] = {
        "passenger": name,
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        "flight_id": flight["id"],
        "source": flight["source"],
        "destination": flight["destination"],
        "fare": flight["fare"],
    }
    used_booking_ids.add(booking_id)
    flight["seats"] -= 1

    print("\nBooking successful!")
    print("Booking ID:", booking_id)
    print("Passenger:", name)
    print("Flight:", flight["id"])
    print("Route:", flight["source"], "to", flight["destination"])
    print("Fare: Rs.", flight["fare"])


def view_booking():
    booking_id = input("Enter booking ID: ").strip().upper()

    if booking_id in bookings:
        print("\n--- BOOKING DETAILS ---")
        for key, value in bookings[booking_id].items():
            print(f"{key.title()}: {value}")
    else:
        print("Booking not found.")


def cancel_booking():
    booking_id = input("Enter booking ID to cancel: ").strip().upper()
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    if booking_id not in bookings:
        print("Booking not found.")
        return

    booking = bookings.pop(booking_id)

    # Restore the seat to the corresponding flight
    for flight in flights:
        if flight["id"] == booking["flight_id"]:
            flight["seats"] += 1
            break

    print("Booking cancelled successfully.")



def main():
    while True:
        print("\n===== FLIGHT RESERVATION SYSTEM =====")
        print("1. View Flights")
        print("2. Search Flights")
        print("3. Book Ticket")
        print("4. View Booking")
        print("5. Cancel Booking")
        print("6. Exit")

        choice = input("Enter your choice (1-6): ").strip()

        if choice == "1":
            display_flights()
        elif choice == "2":
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            search_flights()
        elif choice == "3":
            book_ticket()
        elif choice == "4":
            view_booking()
        elif choice == "5":
            cancel_booking()
        elif choice == "6":
            print("Thank you for using the system.")
            break
        else:
            print("Invalid choice. Please enter 1 to 6.")


if __name__ == "__main__":
    main()
```

```
===== FLIGHT RESERVATION SYSTEM =====
1. View Flights
2. Search Flights
3. Book Ticket
4. View Booking
5. Cancel Booking
6. Exit
Enter your choice (1-6): 1

--- AVAILABLE FLIGHTS ---
ID      Source          Destination     Fare    Seats
F101    Pune            Delhi           Rs.4500  3
F102    Mumbai          Bengaluru       Rs.3500  4
F103    Delhi           Chennai         Rs.5000  2
F104    Pune            Goa             Rs.2500  3

===== FLIGHT RESERVATION SYSTEM =====
1. View Flights
2. Search Flights
3. Book Ticket
4. View Booking
5. Cancel Booking
6. Exit
Enter your choice (1-6): 3
Enter flight ID: F101
Enter passenger name: KETAN

Booking successful!
Booking ID: B1001
Passenger: KETAN
Flight: F101
Route: Pune to Delhi
Fare: Rs. 4500

===== FLIGHT RESERVATION SYSTEM =====
1. View Flights
2. Search Flights
3. Book Ticket
4. View Booking
5. Cancel Booking
6. Exit
Enter your choice (1-6): 6
Thank you for using the system.


...Program finished with exit code 0
Press ENTER to exit console.
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